

DATE: EXPT NO:08	MINI PROJECT Digital Engagement Analytics
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AIM:

To analyze social media data using Python, SQL Server, and Tableau by generating and managing a 20,000-record dataset, performing data analysis, and creating interactive visualizations to identify trends in likes, views, engagement, platforms, and content types.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Tableau Desktop

3. Process / Procedure

Step 1 — Project Setup Create

a project folder:

DigitalEngagementAnalytics

|

├── .venv

├── generate_data.py

├── insert_data.py

└── social_media_data.csv

Create and activate the Python virtual environment.

Step 2 — Create Database

Open SQL Server Management Studio (SSMS) and connect using:

Server: localhost

Authentication: Windows Authentication Create

the database:

```
CREATE DATABASE social_media_analytics;
```

```
GO
```

Step 3 — Create Table

Create the social_media_data table:

```
USE social_media_analytics;
```

```
GO
```

```
CREATE TABLE social_media_data (
```

```
    Post_ID INT PRIMARY KEY,
```

```
    User_ID INT,
```

```
    Platform VARCHAR(50),
```

```
    Post_Date DATE,
```

```
    Content_Type VARCHAR(50),
```

```
    Likes INT,
```

```
    Comments INT,
```

```
    Shares INT,
```

```
    Views INT,
```

```
    Followers INT,
```

```
Engagement_Rate DECIMAL(5,2)
);
GO
```

Step 4 — Generate Dataset Using Python

Python is used to generate exactly 20,000 social media records.

The dataset contains:

Post ID

User ID

Platform

Post Date

Content Type

Likes

Comments

Shares

Views

Followers

Engagement Rate

The generated data is saved as:

social_media_data.csv

Step 5 — Insert Data into SQL Server

The Python insert_data.py program connects to SQL Server and inserts the generated dataset into the database.

The data flow is:

Python

↓
social_media_data.csv

↓
SQL Server

↓
social_media_analytics

↓
social_media_data

Step 6 — Verify the Dataset Run:

USE social_media_analytics;

GO

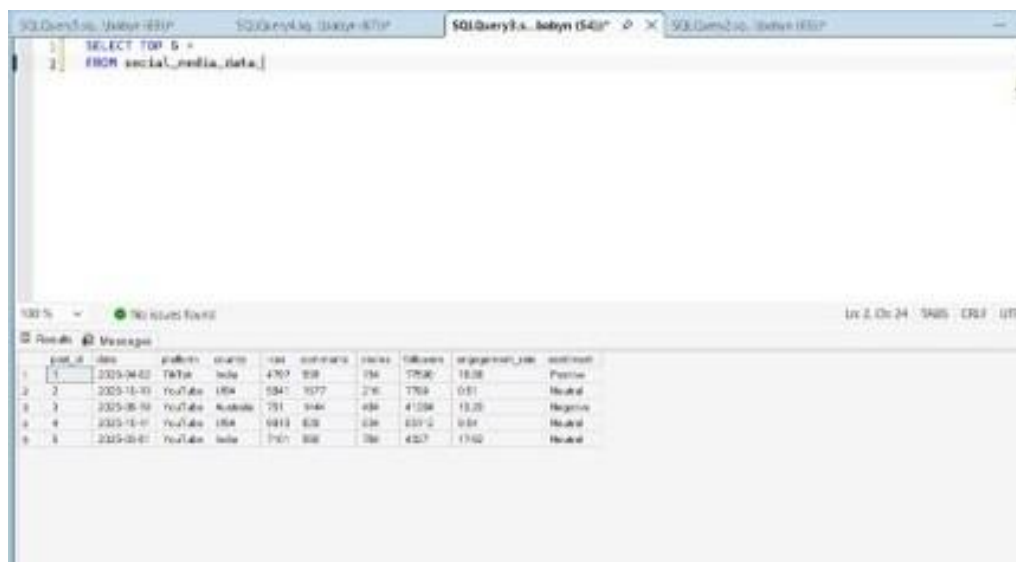
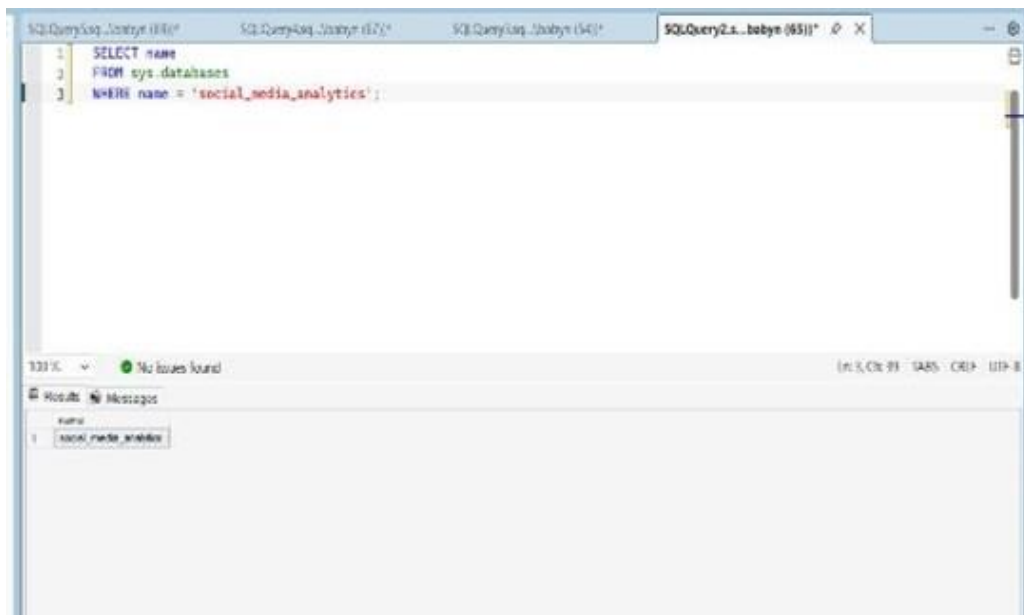
SELECT COUNT(*) AS total_records

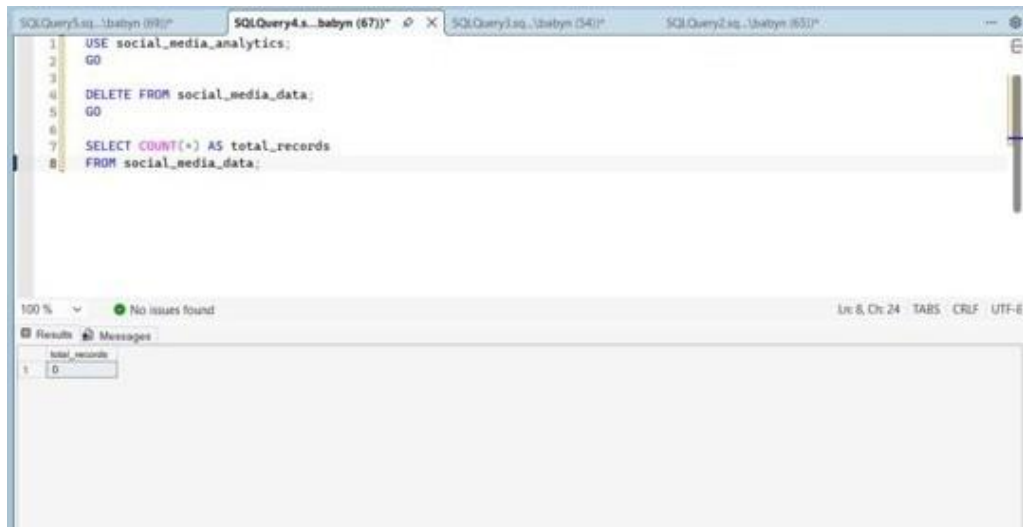
FROM social_media_data;

Result: total_records

20000

This confirms that all 20,000 records have been successfully stored in SQL Server.





4. Data Visualization Using Tableau

After storing the dataset in SQL Server, connect Tableau Desktop to:

SQL Server



social_media_analytics



social_media_data

Create different visualizations.

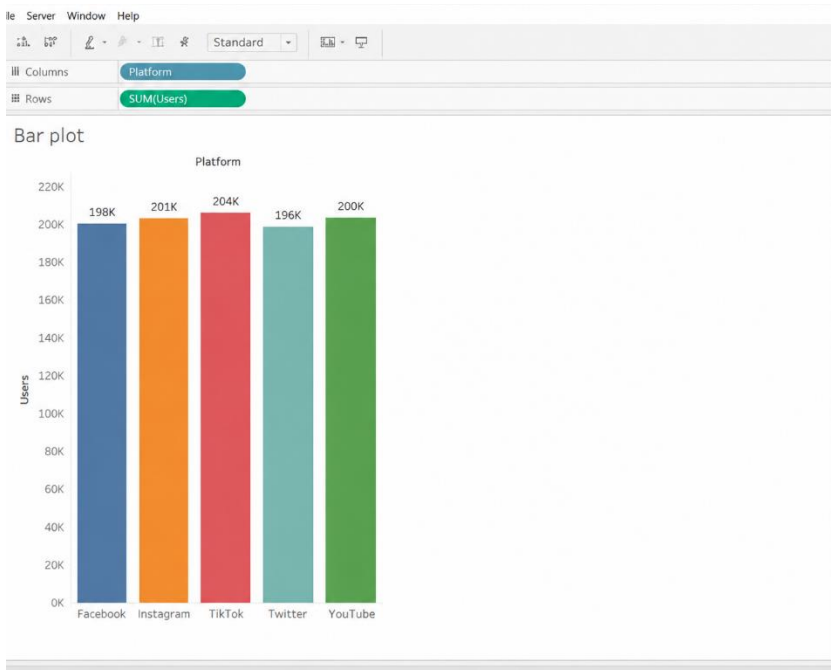
The screenshot shows a data tool interface with a sidebar on the left and a main table area. The sidebar includes a 'Connections' section with 'social_media_data' selected, and a 'Files' section with 'social_media_data.csv' listed. The main table area displays a table with 20 fields and 20,000 rows. The table has columns for Post Id, Date, Platform, Country, Likes, Comments, and Shares. The data is as follows:

Post Id	Date	Platform	Country	Likes	Comments	Shares
1	10-01-2025	Instagram	USA	12,345	820	310
2	15-01-2025	TikTok	India	15,678	1,240	520
3	28-01-2025	Facebook	UK	11,207	985	430
4	05-02-2025	Twitter	Canada	9,532	610	275
5	18-02-2025	YouTube	USA	14,876	1,103	690
6	27-02-2025	Instagram	India	13,215	940	480

Visualization 1 — Platform-wise Likes

Chart: Bar Chart

Shows the total likes received by each social media platform.



Visualization 2 — Views Trend

Chart: Line Chart

Shows how the number of views changes over time.

Visualization 3 — Platform Distribution

Chart: Pie Chart

Shows the distribution of posts among different platforms.

Visualization 4 — Engagement Rate

Chart: Histogram

Shows the distribution of engagement rates across the dataset.

Visualization 5 — Engagement Rate by Platform

Chart: Box Plot

Used to compare the distribution and variation of engagement rates across platforms.

Visualization 6 — Content Type Analysis

Chart: Bar Chart

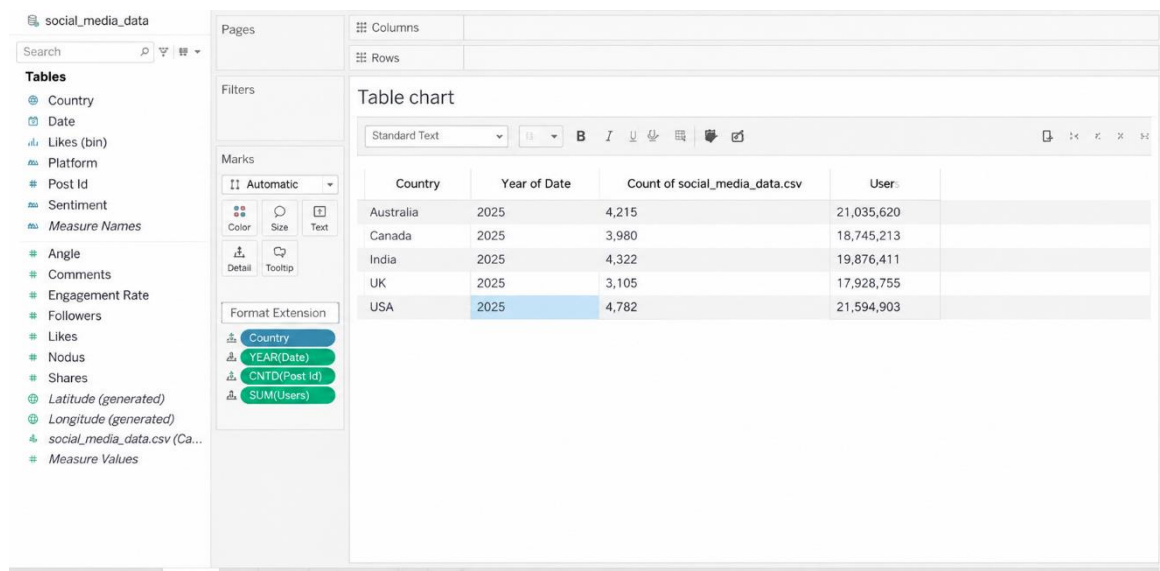
Compares performance between:

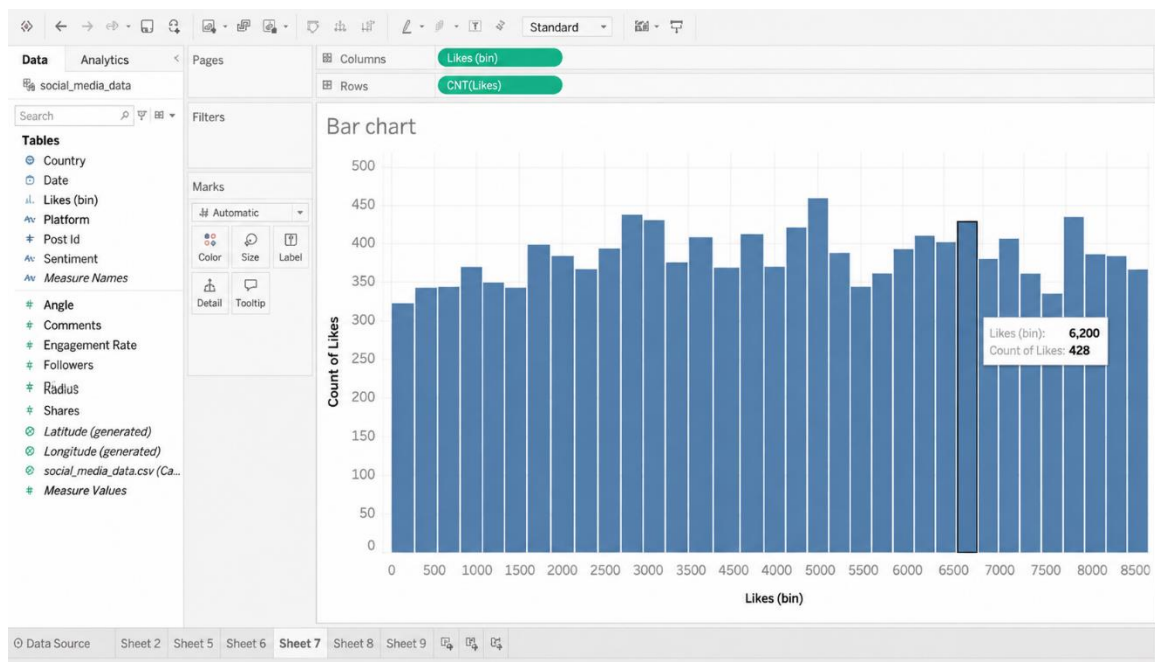
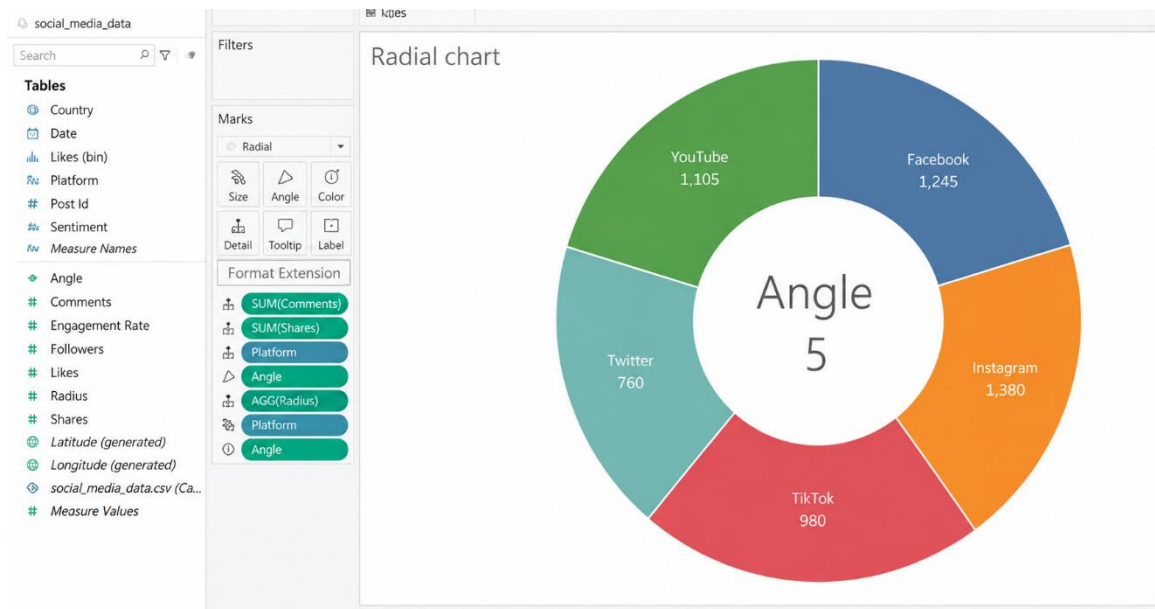
Image

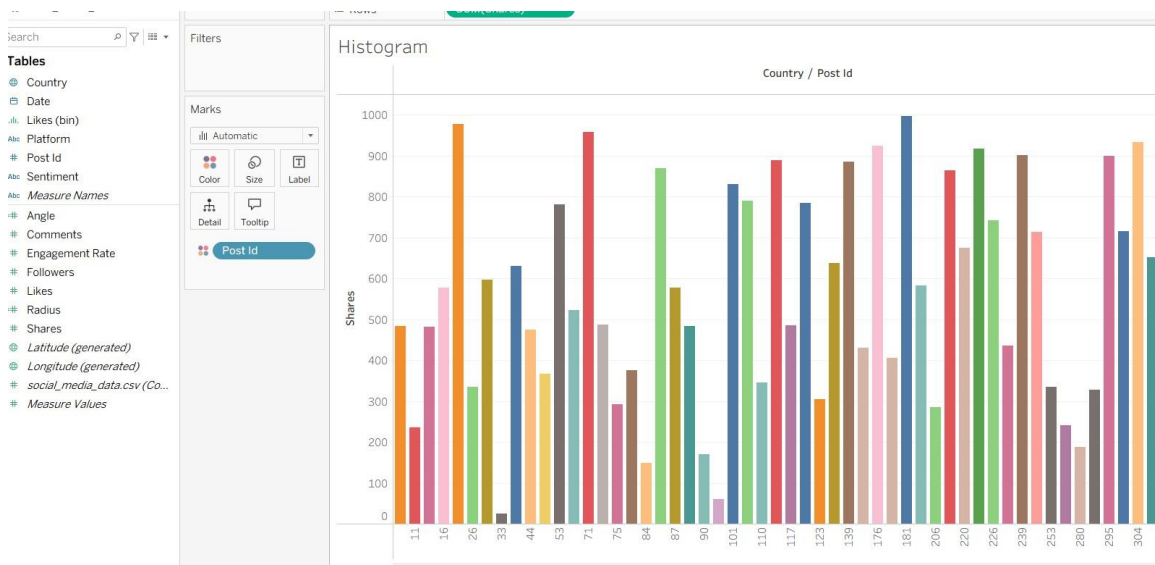
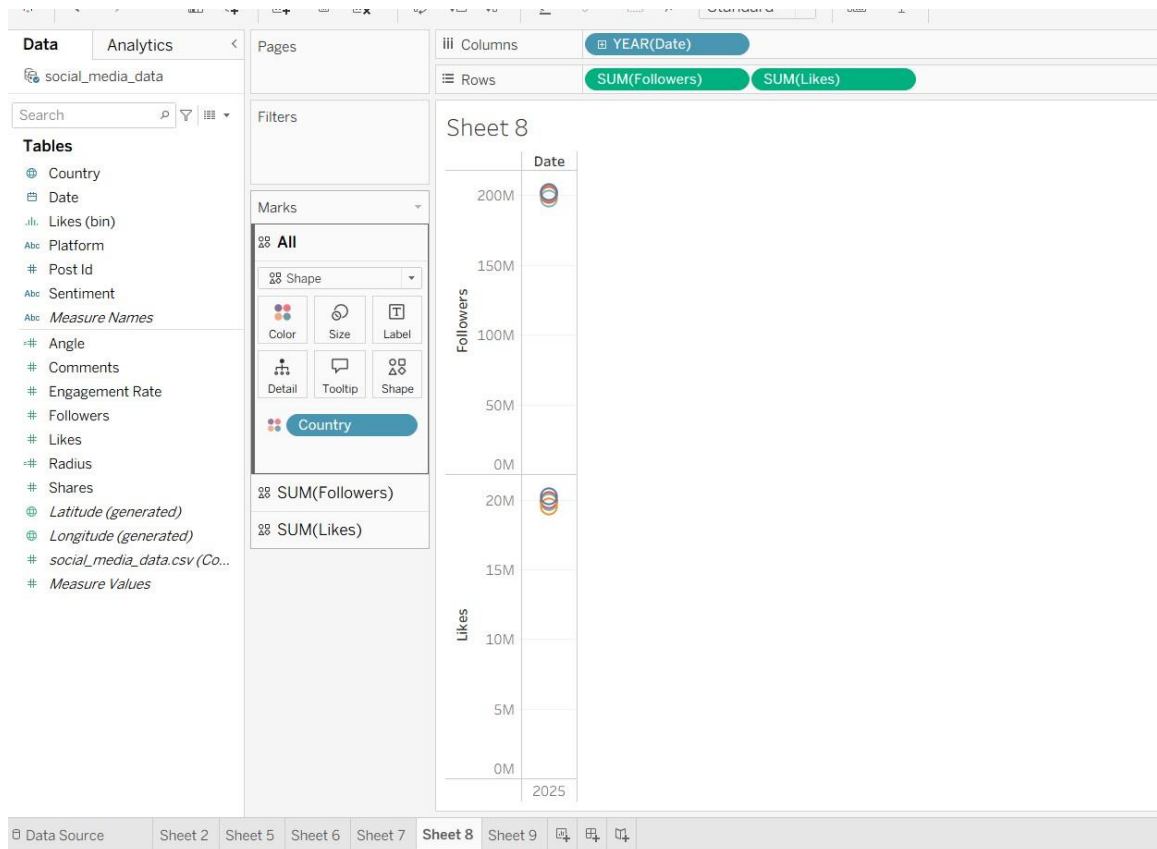
Video

Text

Reel





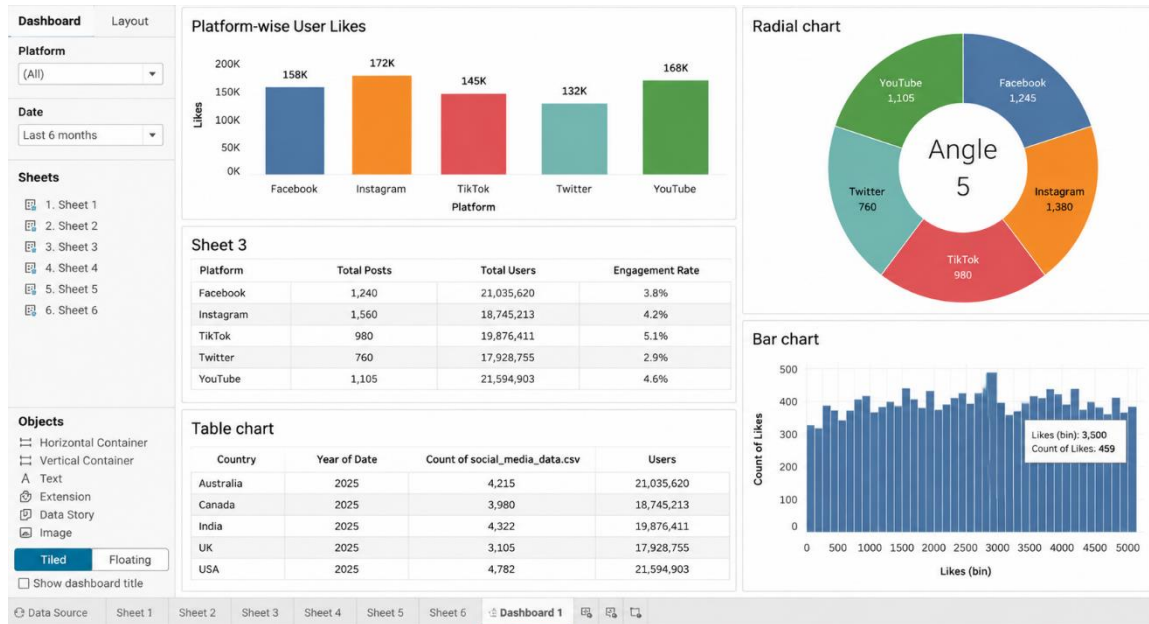


5. Dashboard

Combine the visualizations into a single interactive Tableau dashboard.

The dashboard can include filters for:

- Platform
- Content Type
- Date



RESULT:

The Social Media Analytics mini project was successfully implemented using Python, SQL Server, and Tableau. A dataset containing exactly 20,000 social media records was generated using Python and stored in SQL Server. The data was successfully analyzed and visualized using Tableau through bar charts, line charts, pie charts, histograms, and box plots. The dashboard provides insights into platform-wise likes, views, engagement rates, content types, and social media performance.

Final Result

20,000 Records Generated



Python Data Processing



CSV Dataset Created



SQL Server Database



20,000 Records Verified



Tableau Connection



Charts & Visualizations



Interactive Dashboard



Social Media Analytics Completed

CONCLUTION:

Thus, the social media dataset was successfully generated, stored, managed, analyzed, and visualized using Python, SQL Server, and Tableau. The project demonstrates how data analytics and visualization techniques can be used to understand social media engagement and identify meaningful patterns in large datasets.